THE MILLENNIUM PROBLEMS

BY

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WORKBOOK
EVERY CLAIM, WHAT IS BEHIND IT, AND WHAT IT WANTS

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1

NAVIER–STOKES: THE TERMS EQUATION (1) LEAVES OUT

Purpose: Hold every claim of the first observation of the thought experiment beside what is behind it. A reader then sees at a glance which claims are proved, which were read, which arrived by report, and which are still only theory. **Scope:** the thought experiment recorded on 2026-10-01 as a run of questions and answers, and the research paper chapter *The terms equation (1) leaves out*, whose propositions this chapter names by number.

1.1 HOW AN ENTRY IS RETAINED

Every claim carries one status, and the status says what backs it.

Status	What it means
proved	a proof is written out in the research paper and the number of its proposition is given
read	a source was opened and the claim is in it, in the words given
by report	the claim reached this work through another source, and that source was not opened
theory	stated, argued from scaling or by analogy, and not proved
not so	fails on its own terms; retained, with the reason
open	neither proved nor displayed false here

A claim changes status only when a proof or a reading changes it, and the change is recorded beside it. Nothing is deleted. A claim that failed stays, with what failed it.

1.2 THE FIRST OBSERVATION

The thought experiment opens by asking where the Navier–Stokes equations are bound: where a value is assigned to the fluid instead of being asked of it. Its answer is that they are bound in nearly every term. Density is assigned, viscosity is assigned, temperature is left out, and the fluid is put in a box before anything is known about its container. Each of these is known from measurement to be wrong for a real fluid, or is a choice made before the fluid was looked at.

The program it sets is to write every term in vector form and to hold nothing fixed that the fluid has not been asked for. The fluid fills all of space until a container is measured, and the container

is read from the motion, through spherical harmonics of the wave patterns the motion carries, and not assumed.

The same shape runs through the whole set of equations: put the fluid in a box, approximate what it does there, and average the temperature over a volume. Each step drops information the motion carries. This chapter enters the dropped terms and the box. Averaging over a volume is the second observation and has the next chapter. Every other step of that shape goes in a later one.

The record goes on to entropy, observation and arithmetic. Those parts bear on no claim in this observation and are not entered here.

1.3 ENTRIES

Claim	Status	What is behind it
Equation (1) is bound: its constants are assigned to the fluid and not asked of it.	read	Fefferman's (1) carries one constant, ν, and no density and no temperature.
Density varies with temperature, and a constant density is assigned instead of measured.	proved	Proposition 4: with $\rho'(T) \neq 0$, condition (2) forces conduction to cancel shear heating at every point. Silent where ρ' vanishes.
Viscosity is not a constant, and pulling μ out of the divergence drops a term.	proved	Propositions 1 and 3: the dropped term is $2\mu'(T)\, D\nabla T$, zero only where ∇T lies in the null space of D.
The fluid defines its own state if it is let.	proved	Proposition 2: any viscous motion that decays at infinity makes its own temperature gradient.
With vector calculus the dropped terms cannot be hidden.	proved	Proposition 1: the shear part is twelve terms per component and one product in vector form.
Friction is not linear, anywhere.	open	The dependence measured in the research paper is on temperature, -2.45 percent per kelvin for water at 20 °C, evaluated from IAPWS R12-08 [10]. A dependence of water's viscosity on $\lvert D\rvert$ was not measured. Read since: Stokes reports from the pipe experiments of his time that the tangential force at a wall varies nearly as the square of the velocity, except at small speeds [20], and Navier takes the linear law as a principle. Neither measures water's viscosity against $\lvert D\rvert$, and the status stays open.

Claim	Status	What is behind it
Put a vector magnitude where a value is not known.	read	A viscosity $\mu(\lvert D\rvert) = \mu_0 \lvert D\rvert^{p-2}$ is that move. For $p > 11/5$ in three dimensions a solution obeying the energy equality is obtained; for $p = 2$ that is not known. Read in Lienstromberg, Schiffer and Schubert [16].
Never put the fluid in a box unless it is in one; a box destroys information.	proved	Proposition 6: the box retains the transform of the field on the integer lattice and drops the rest, and nonzero flows with zero divergence enter it as zero. It loses nothing when the fluid already fits in one cell.
The box sets directions the fluid does not have.	proved	Proposition 7: 48 rotations and reflections about a point act on the box, against an infinite group on all of space.
Spherical harmonics of the wave patterns recover the container.	theory	Stated in the thought experiment. Not proved, and no computation was run.
The forced construction breaks down under real conditions.	proved, in part	Proposition 5: the rate of shear heating has no bound at the singular time, and its total is finite. That the breakdown would not happen with the coupling retained is open.
The forced construction failed to prove the theorem.	not so	The theorem is about equation (1), and its proof was not read. Nothing here bears on whether the proof is correct. What the propositions bound is what the theorem is about. Read since, at the level of its statements, its order of choices and its closing estimates; nothing read bears against it [17].
A collapsing vortex in water vaporizes near 0.7 nanometers of width, found by independent work.	by report	Reached this work only through the thought experiment. No source was located or opened, and nothing rests on it. Since: the nearest source found puts the first failure in water at cavitation near a millimeter of core radius and gives 0.3 nanometers as the molecular scale; its 0.7 is in micrometers and for air [4]. It bears on the claim and does not support it.

Claim	Status	What is behind it
An unbounded value is a sign that an assumption has left the fluid.	theory	Proposition 5 is the proved part: the unbounded factor is a factor of the dropped term. The scaling entry below is the rest. Read since: of a light-emitting bubble, "since the light emission occurs as a (mathematical) singularity is forming, any transport process which a theorist chooses to incorporate in this model can affect the result" [1]; and of the forced construction, a real fluid leaves the description of the equations long before the singularity [4].

1.4 A SCALING ESTIMATE, MARKED AS THEORY

Take a collapsing core of radius r and speed U, with $|D| \sim U/r$. Suppose the shear heat in the core, of order $\mu U^2/r^2$ per unit volume, is carried off by conduction across the core, of order $k\,\Delta T/r^2$. Then the temperature excess of the core is

$$\Delta T \sim \frac{\mu U^2}{k},$$

independent of r. The ratio of the dropped term to the retained viscous term is of order $|2D\nabla\mu|/|\mu\Delta u| \sim 2\,|(\ln\mu)'|\,\Delta T$, since both carry one more factor of $1/r$ than D. The ratio is therefore

$$2\,|(\ln\mu)'|\,\frac{\mu U^2}{k},$$

and it passes one at

$$U^* = \sqrt{\frac{k}{2\mu\,|(\ln\mu)'|}}.$$

Any collapse in which U grows without bound passes U^*. Below it the dropped term is a small error. Above it the dropped term is as large as the term retained.

Why this is theory and not proved. It assumes one length scale for the core, a balance between shear heating and conduction that holds at every moment, and derivatives estimated by size alone. None of the three is displayed.

For water at 20 °C and atmospheric pressure, IAPWS R15-11 [11] gives $k = 0.5980$ watts per meter per kelvin and IAPWS R12-08 gives $\mu = 1.0016$ mPa s with $(\ln\mu)' = -0.0245$ per kelvin, and $U^* = 110$ meters per second. At 0 °C it is 67 meters per second, and at 40 °C it is 160. The number is computed. The estimate it comes from is still theory.

Read since. Duraiswami [4] puts the axis of the forced construction in water at cavitation when the swirl reaches 10 to 17 meters per second, at a core radius near a millimeter, with the viscous heating over the rest of the collapse near 0.01 kelvin. On those numbers water cavitates

well below U^{*}, and the dropped term is not the first assumption to fail in water: the pressure floor is. The estimate stays theory, and it now has a competitor that fails first.

Gruntfest and Becker [8] find that shear at a fixed stress between walls retained at a fixed temperature, in a liquid whose viscosity falls exponentially with temperature, has no steady state above a peak stress. Water at 20 °C reaches that peak when the walls move at 666 meters per second, $6.04\,U^{*}$. With the walls moving at a fixed speed the steady layer exists at every speed, and its middle is at T_c with $V^2 = 8\int_{T_0}^{T_c} k/\mu\, dT$, independent of the gap. On the IAPWS values the middle reaches 100 °C only near 956 meters per second, and at the 17 meters per second of cavitation in the forced construction it is 0.06 kelvin above the walls.

1.5 WHAT IT WANTS

Every term this chapter enters is collected, with the others, in the table of terms to put back in the last chapter.

The estimate above wants a computation: the forced construction, or any collapsing solution of (1), with T carried beside it through the energy balance, and $2\mu'(T)\,D\nabla T$ evaluated along the way and compared with $\mu\Delta u$. That replaces the scaling with a number taken on an actual solution. It is not built.

The gap in Proposition 3 wants a proof either way: whether a flow can hold its own temperature gradient inside the null space of its own rate of strain at every point and time.

The spherical harmonic entry wants a statement precise enough to prove: what feature of a container, read from which harmonics of the motion on all of space, and with what loss.

2

NAVIER–STOKES: WHAT THE AVERAGE RETAINS

Purpose: Hold every claim of the second observation of the thought experiment beside what is behind it, retained the same way as the first. **Scope:** the second observation, and the research paper chapter *What the average retains*, whose propositions this chapter names by number.

2.1 THE SECOND OBSERVATION

Take a quintillion molecules of a fluid. Each one feels the field and every neighbor. Reducing what they do to a few numbers per point looks absurd once that is in view. The observation does not say the average is wrong. It says the average over enough volume and enough time gives an entropic equivalence that is approximate and not exact, and that the equations are built on that average.

2.2 ENTRIES

Claim	Status	What is behind it
The field is continuous and the molecules sit in it; the differences of potential between any two are uncountably many.	proved	Proposition 8: a continuous field takes every value between its values at two molecules, and an interval of real numbers is uncountable.
Normalized to 0 and 1, every one of those values is retained.	proved	Proposition 8: $t \mapsto t/(1+t)$ carries $[0,\infty)$ one to one onto $[0,1)$, retention order, and cuts none off.
The continuum of the first observation and the molecules of the second pull in opposite directions, and the research paper must take one.	not so	Put forward while this observation was entered, and taken back. The field is defined at every point and the molecules are a finite number of points in it. Both hold at once. Proposition 8.
Equation (1), read as a law of a real fluid, is a law of averages over many molecules.	proved	The research paper reads it so in its second chapter. Equation (1) carries no molecules and holds a few numbers per point.

Claim	Status	What is behind it
The average throws away what each molecule is doing.	proved	Proposition 9: a cell of N molecules retains four numbers, and every state in a set of dimension $6N - 4$ gives the same four.
The average is an entropic equivalence.	proved	Proposition 9: the set of states with the same cell values is the equivalence class, and its size is what the Boltzmann entropy measures.
Over enough volume the equivalence is approximate and never exact.	proved, under independence	Proposition 10: the scatter of the average is $s/\sqrt{N}$, positive for every finite N. A liquid is not independent, and the rate is then the best case.
Over enough time the same holds.	theory	The time form, with the time over which the motion stays correlated in the place of one molecule, is stated and not proved.
A solution that breaks down leaves too little volume and too little time to average over.	proved	Proposition 11: in the class of the forced construction, the length ℓ of the flow and the time ℓ^2/ν have no positive lower bound.
In water, that happens at lengths a solution passes through.	computed	33.4 molecules in a cube of side one nanometer at 20 °C, with a scatter of 17 percent under independence, and ℓ^2/ν of one picosecond there. Density from IAPWS R6-95 (2018) [12], viscosity from IAPWS R12-08 [10], both evaluated by programs that reproduce their verification tables.

The status *computed* is added here: a number worked out from stated inputs, with the inputs named and their source given.

2.3 WHAT IT WANTS

The time form of Proposition 10 wants a proof under a stated rate at which the motion stops being correlated with itself.

The entries on water want the collision and correlation times of liquid water, read from a source, set against the times of Proposition 11. Where those cross is where the average stops being available, and that number has not been taken.

The count of Proposition 9 wants extending to the field between the molecules: what an average over a cell retains of a continuous field, and what it drops.

3

NAVIER–STOKES: TERMS TO PUT BACK

Purpose: Collect every known member of P_{all} and F_{all} that equation (1) drops, with the term that puts it back and the source that shows it present; name what is not known; and look across the known members for a pattern left over. **Scope:** the research paper chapters on the dropped terms, the average and the two sets, whose propositions this chapter names by number, and the sources registered with them.

3.1 THE METHOD

The approach to the properties of a fluid has three steps. Examine everything known. Examine what is not known, and name it as not known. Then look across the parts that are known and see whether any pattern is left. The three sections below are those steps in order.

P_{all} is the set of all properties of a fluid and F_{all} the set of all forces a fluid can experience, known and not yet known, and neither list is claimed complete. Equation (1) retains $P_{(1)} = \{\nu\}$ and $F_{(1)} = \{-\nabla p,\ \nu\Delta u,\ f\}$.

3.2 WHAT IS KNOWN

Each row is a member displayed present in a real fluid and dropped by equation (1). The status column uses the status names of the earlier chapters, and *proved here* marks a proof short enough to be written in the row itself.

Member	What equation (1) does	Term to put back	What shows it present
In P_{all}			
Viscosity that varies with temperature	holds ν constant	$2\mu'(T)\,D\nabla T$ in the momentum balance; Propositions 1 and 3	computed: −2.45 percent per kelvin at 20 °C, IAPWS R12-08 [10]
Viscosity that varies with pressure	holds ν constant	$2(\partial\mu/\partial p)\,D\nabla p$	computed: −0.031 percent per megapascal at 20 °C, IAPWS R12-08 [10]; the sign turns positive between 25 and 40 °C

Member	What equation (1) does	Term to put back	What shows it present
Density that varies with temperature	sets $\rho = 1$	$\rho(T)$ in the inertia, and $\rho'(T)\,DT/Dt + \rho\theta = 0$; Proposition 4	computed: expansion of 2.07×10^{-4} per kelvin at 20 °C, zero at 3.978 °C, IAPWS R6-95 (2018) [12]
Density that varies with pressure: the weight of the fluid above, and stirring	sets $\rho = 1$	$\rho(p, T)$ in the inertia and the mass balance, with the compressibility $\partial \ln \rho / \partial p$	computed: 4.589×10^{-10} per pascal at 20 °C, a column at rest denser at the bottom by 4.49×10^{-6} per meter, and a turning cup less dense at the bottom center and denser at the bottom wall; IAPWS R6-95 (2018) [12], intuition I6
A finite speed of sound	condition (2) makes it infinite; Proposition 12	$\theta \neq 0$ and an equation of state $p(\rho, T)$	computed: 1482.35 meters per second at 20 °C, IAPWS R6-95 (2018) [12]
Bulk viscosity	removed with θ by condition (2)	$(\lambda + \mu)\nabla\theta + \theta\nabla\lambda$, with $\lambda + \frac{2}{3}\mu$ the bulk viscosity	read: about three times the shear viscosity from 7 to 50 °C, 2.469×10^{-3} Pa s at 25 °C; Holmes, Parker and Povey [9]
Heat: a temperature field, its conduction, and the heat shear puts in	holds no temperature	$\rho c\, DT/Dt = \nabla \cdot (k\nabla T) + 2\mu\lvert D\rvert^2$; Propositions 2 and 5	computed: $k = 0.5980$ watts per meter per kelvin at 20 °C, IAPWS R15-11 [11]
A level of pressure at which the liquid breaks	holds no level of pressure; Proposition 13	pressure fixed in level, and an equation of state that holds the phase boundary as a free surface; the coupled system of the research paper, Propositions 14 to 17	read: −168 MPa from nucleation theory at 20 °C, about −140 MPa in inclusions, a few to a few tens of megapascals below zero by other methods; Caupin and Herbert [2]
Molecules, and the field between them	averages both over a cell; Propositions 8 to 11	none written yet	proved, and computed: 3.337×10^{28} molecules per cubic meter at 20 °C, IAPWS R6-95 (2018) [12]
In F_{all}			
Thermal fluctuating stress	holds none	$\nabla \cdot \tilde{\tau}$, Gaussian with covariance $\frac{2\nu k_B T}{\rho}(\delta_{ik}\delta_{jl} + \delta_{il}\delta_{jk} - \frac{2}{3}\delta_{ij}\delta_{kl})\,\delta^3(x - x')\,\delta(t - t')$	read: the scale at which it matters is about the Kolmogorov length; Eyink, Bandak, Goldenfeld and Mailybaev [5]

Member	What equation (1) does	Term to put back	What shows it present
Light, from energy concentrated in a collapse	holds no channel for it	a loss of energy to radiation in the balance of energy	read: flashes shorter than 50 ps from collapsing bubbles, energy concentrated by twelve orders of magnitude, brightness up a hundredfold from 40 to 0 °C; Barber, Hiller, Lofstedt, Putterman and Weninger [1]; intuition I11
Gravity acting on a density that varies	takes f from outside, and $f = 0$ in statements (A) and (B)	$\rho(T)\, g$	proved here: $\nabla \times (\rho g) = \nabla\rho \times g$, nonzero wherever ∇T is not parallel to g and $\rho'(T) \neq 0$, and a force with nonzero curl cannot be taken into $-\nabla p$
The force of statements (C) and (D)	admits any smooth f with the decay of condition (5)	none: the want is to name the member of F_{all} that f is for	open

3.3 WHAT IS NOT KNOWN

Members not yet measured. By the definition of the two sets, they cannot be listed. Their place in this chapter is the statement that the table above is not the whole of either set.

Intuitions not yet tested. Each is retained in the author's words, with the source that would test it and what would count as a hit or a miss, in the log of intuitions recorded before reading that sits beside this workbook. An entry there is scored only after the source is read.

Candidates with no obtained source. Memory in the stress, gas dissolved in the liquid, and forces at a free surface are candidates. None is entered until a source is obtained and read.

Questions the earlier chapters leave open. Whether the viscosity of water depends on $|D|$ at rates a solution reaches. Whether a flow can hold its own temperature gradient inside the null space of its own rate of strain. The time form of Proposition 10. Whether spherical harmonics of the motion recover the container. Whether retention any member above changes whether solutions break down: that is open for every row.

An analogy, retained as theory. Microcavitation is to water what quantum foam is to space: a fluid under so much tension that it must move angular momentum between the bulk and a boundary, and it can do so only through heat. The tension is in the fluid itself and not in a surface.

Two parts of this are read. Caupin and Herbert [2] describe a liquid stretched below its saturated vapor pressure, with a pressure that can go negative, and read a negative pressure as a density reduced from the equilibrium one: the fluid itself is pulled apart. They describe homogeneous nucleation as the birth of the new phase by thermal fluctuations, at a rate proportional to $\exp[-E_b/(k_B T)]$, over a barrier $E_b = 16\pi\sigma^3/\big(3(P_\sigma - P)^2\big)$. The tension of the fluid is the factor $P_\sigma - P$, and the more of it the lower the barrier. The surface tension σ enters only as the cost of the wall of the bubble once one forms, and it is a further member of P_{all} that equation (1) does not hold. The break is crossed by heat.

The analogy goes further. Introducing space into the fluid sets off a chain, the energy available compounds for a short while and grows exponentially, and the end of the chain is the microcavitation. With $\Delta P = P_\sigma - P$ the tension and $A = 16\pi\sigma^3/(3k_BT)$, the links are these.

1. Space is introduced: the fluid is stretched and its density falls below the equilibrium one. *Read*, Caupin and Herbert.

2. The pressure falls below the saturated vapor pressure and can go negative: the fluid is under tension ΔP. *Read*, the same.

3. The barrier to a bubble falls as the tension rises, $E_b = A\,k_BT/\Delta P^2$, as the inverse square of the tension. *Read*, their equation (2).

4. The rate of nucleation, proportional to $\exp(-A/\Delta P^2)$, grows exponentially:

$$\frac{d}{d\,\Delta P}\,\ln(\text{rate}) = \frac{2A}{\Delta P^3} > 0,$$

   and climbs from negligible to certain over a narrow range of tension. *Proved here* from their equation (2).

5. Heat carries one fluctuation over the barrier: a bubble reaches the critical radius $R_c = 2\sigma/\Delta P$. *Read*: nucleation is the birth of the new phase by thermal fluctuations.

6. Past R_c the work to form a bubble, $-\frac{4}{3}\pi R^3\Delta P + 4\pi R^2\sigma$, falls as R grows, and the bubble grows by itself: the microcavitation. *Proved here* from their equation (1).

Every link from the stretching to the microcavitation is read or follows from the review's own equations.

The chain as a seiche, retained as theory. Conceptually the chain is the same as a seiche: energy aligned. A seiche is a located wave in a basin, and its energy builds where the motion arrives in phase. One part of this is read. Caupin and Herbert report that the acoustic experiments reaching the max tension in water used aligned energy: Galloway a located wave in a spherical resonator, reaching −20 MPa in water degassed to 0.02 percent saturation; Greenspan and Tschiegg a located wave focused in a steel cylinder, reaching −16 to −21 MPa at 30 °C; and their own experiment a few cycles of a 1 MHz wave tightly focused at the center of a hemisphere, with an effective time of 45 nanoseconds. In each, the negative swing of the aligned wave supplies the first link of the chain. That the chain and a seiche are the same in concept is not in any source read, and no source on seiches is obtained. How long the compounding lasts in time, and what each link costs in a flow of equation (1), are not in any source read. The transfer of angular momentum between bulk and boundary, and the resemblance to quantum foam, are not in any source read either, and are not displayed.

3.4 THE PATTERN LEFT OVER

Nine of the thirteen rows carry temperature. Viscosity, density, the speed of sound and bulk viscosity each vary with it in the sources read. Heat is its balance. The level at which water breaks moves with it in the review read, and the light of a collapsing bubble grows a hundredfold as the

water is cooled from 40 to 0 °C. The fluctuating stress is proportional to $k_B T$, and gravity reaches the motion through $\rho(T)$. The four that do not are the two dependences on pressure, of viscosity and of density, the molecules, and the external force. Pressure is the second variable the rows share, and equation (1) holds pressure only as a gradient with no level, Proposition 13.

Equation (1) has a place for every variable here except temperature, and Proposition 2 shows that a viscous flow makes its own temperature gradient. The members equation (1) drops are therefore mostly members the flow itself moves. That is the pattern left after the known parts are positioned side by side.

Where the pattern leads. The fluids of the world are transitive: every one of them carries the members of the table, whatever its scale and however it is stirred. Along every route to an unbounded value that this work has examined, one of those members acts first: the molecules, thermal noise, the heat of shear, or the level at which the fluid breaks. A real fluid therefore leaves the description of equation (1) before it could reach the value, and no singularity is expected in observation, though other extreme objects in fluids are observed. Intuition I9 in the log beside this workbook scores this: the first part as a hit for the routes examined, and the absence of observation as open until a source on it is obtained.

Status: theory. The pattern is read off this table. It is not a law, it rests on the rows entered so far, and a row added later can break it.

The counter argument, retained as theory. In the author's words: "Just like with every other high energy reaction, the forces need to stay coupled until criticality." The forced construction reaches its singular time by retention two families of pulses aligned, so that their averaged fluxes feed the core all the way down. A real fluid stays coupled that way only while it stays liquid, its flow stays near laminar, and its viscosity and density stay what equation (1) holds them at. Proposition 19 of the research paper proves that none of those can be decided in equation (1): every solution of it, carried with a temperature, is compatible with a fluid past its vapor point everywhere.

The other half is the heat of the core. Propositions 15 and 18 set the temperature that shear holds at a boundary by the square of the velocity difference across it. Along the forced construction the peak swirl grows as $\tau^{-1/2-h}$ (Duraiswami [4], Table 7). Proposition 20 of the research paper proves that a column turning with a swirl of that form, its heat leaving by conduction alone, has a temperature on its axis that grows as τ^{-1-2h}, the square of the peak swirl, and it gives that temperature as an exact integral. Proposition 21 gives the part the outer swirl makes at $h = 0$ in closed form, through E_1 and e^{-s} at $s = \mathrm{Pr}$. The temperature on the axis has no bound, while a liquid turns to vapor at a finite temperature, and in water the tension route of Proposition 17 lets go sooner still, at 10 to 17 meters per second. The column has no inflow and its swirl is given. This is proved for the column and retained as theory for the construction itself.

3.5 REPRODUCING THE MATCHED CORE

Why the whole core is needed. The column of Proposition 20 moves only by its swirl. The matched core of Duraiswami [4], Section 7, also carries an axial velocity $U(X, \eta)$ and a radial

inflow. In his variables $u_\theta = q^{-A}\sqrt{2X}\,F$, $u_z = q^{-A}U$ and $X = r^2/(2q)$, with $\nu = 1$ and $A = 1/2 + h$, the radial derivative is $\partial_r = (r/q)\,\partial_X$ and $r^2 = 2qX$. Then

$$\big(r\,\partial_r(u_\theta/r)\big)^2 = q^{-1-2A}\,(2XF_X)^2, \qquad (\partial_r u_z)^2 = q^{-1-2A}\,2X\,U_X^2.$$

The shear of the axial velocity heats the fluid at the same power of q as the shear of the swirl, and Proposition 20 holds only the second. The strain of the radial velocity enters at q^{-2}, smaller by the factor q^{2h}, which is 0.66 at $h = 1/100$ and $q = 10^{-9}$. The inflow and the axial flow also carry heat along the axis, and the column has neither. The temperature on the axis of the construction is therefore not the integral of Proposition 20 with his F put in it. It needs his whole core, and the core is reproduced first.

Proved here. For the swirl, $u_\theta/r = q^{-A-1/2}F$ and $r\,\partial_r = 2X\,\partial_X$. For the axial velocity, $\partial_r u_z = q^{-A}U_X\,r/q$, whose square is $q^{-2A}U_X^2\,2X/q$.

The series stage. The core is the Cauchy problem in X of the leading-order system of OpenAI [17], eq. (4.13), solved from analytic data on the axis by the recursion of Duraiswami's eqs. (17) and (18). The driver `axis_series` in `examples/navier_stokes` holds each Taylor coefficient as an exact polynomial in η over a power of $1 - 2h\eta^2$, with no grid in η and no filter. On his test data $F_0 = 1, U_0 = \eta/2, \Pi_0 = 0$, $h = 1/100$:

Quantity	Duraiswami	Exact, this work
Residuals of the sum to $k = 40$ on $0 \le X \le 1.17$	below 10^{-10}	4.6×10^{-19} (θ), 3.8×10^{-17} (z)
Residuals of the sum to $k = 80$		8.1×10^{-40} (θ), 2.1×10^{-37} (z)
Parity of F_k and U_k in η	even and odd to 10^{-12}	even and odd exactly
Radius of convergence in X	3.9 to 4.0	from F at $k = 80$: ratio 4.0173, root 4.0059
		from U at $k = 80$: ratio 3.901

The size of a coefficient here is its largest magnitude at $\eta = j/16$, $j = -16$ to 16; his is the largest at his collocation nodes.

Status: computed. The residuals and the parity reproduce his. The estimates from F lie above his range by 0.1 to 0.4 percent, and the estimate from U lies inside it.

The pressure datum. His join fixes the axis value of the pressure by

$$\Pi_0(\eta) = G(\Pi_0)(\eta) = -\int_0^{X_b} F_{\mathrm{blend}}[\Pi_0](X,\eta)^2\,dX + \Pi_{\mathrm{ext}}(X_b,\eta),$$

where the core, and with it the blended swirl, depends on Π_0 through $Z_{-2A}\Pi$. Past the join the swirl is the heat exterior of OpenAI's eq. (4.29), and with $H(Z) = Z^{-1-h}\,U(1+h,2,1/Z)$, U the confluent hypergeometric function of Tricomi, it is $F_{\mathrm{ext}} = (c/\sqrt{2})\,(2d)^{-1-h}\,U\big(1+h,2,X/(2d)\big)$, a solution of Kummer's equation in X. The right side is transcendental in η, while the exact series takes Π_0 only as a polynomial. The datum is represented as a polynomial $\bar p$ with rational coefficients together with a proved ball around it, by the following.

Lemma (proved here). Let B be a Banach space, $\Phi : B \to B$ continuously differentiable, $\bar p \in B$, and $\mathscr{A}$ a bounded injective linear map on B. Suppose $\|\mathscr{A}\Phi(\bar p)\| \le Y$ and $\|I - \mathscr{A}\,D\Phi(p)\| \le Z < 1$ for every p in the closed ball of radius r about $\bar p$, with $Y + Zr \le r$. Then Φ has exactly one zero in that ball.

Proof. Put $T(p) = p - \mathscr{A}\Phi(p)$. On the ball, which is convex, $\|DT\| \le Z$, and T has Lipschitz constant Z. For p in the ball, $\|T(p) - \bar{p}\| \le \|T(p) - T(\bar{p})\| + \|T(\bar{p}) - \bar{p}\| \le Zr + Y \le r$. The map T sends the ball into itself and contracts it, and has exactly one fixed point there. A fixed point of T is a zero of $\mathscr{A}\Phi$, and $\mathscr{A}$ is injective. □

Here $\Phi(p) = p - G(p)$ and $\mathscr{A}$ approximates the inverse of $I - DG(\bar{p})$. The fixed point is repelling, as Duraiswami reports. The plain iteration therefore cannot be used, while the lemma does not need it.

Not established. The space. G takes Π_0 through $\partial_\eta \Pi_0$ inside $Z_{-2A}\Pi$, and each order of the series in X takes one more derivative in η. G is not bounded on continuous functions of η. The ball is to be taken among functions analytic on an ellipse about $[-1, 1]$, where the loss of a derivative at order k is paid by the factor $1/(2(k+1)^2)$. Which ellipse G maps into itself, and the bounds Y and Z on it, are not computed.

Status: theory. The lemma is proved. Its use for the datum waits on the space. The right side of the datum at one η is computed as an exact form: rational coefficients times rational powers of e and symbolic terms, E_1 and the Kummer values among them.

3.6 WHAT A LABORATORY WOULD SHOW

Propositions 14 to 17 of the research paper put the first boundary water reaches at the cavity. From atmospheric pressure at 20 °C the axis of a turning column reaches the vapor pressure at a swirl near 7.6 meters per second, and shear warms a layer moving at the same speed by about a hundredth of a kelvin. Duraiswami [4] proposes the apparatus: a swirl chamber between porous walls driven toward collapse, with the onset of cavitation on the axis and the loss of the steady state as what is observed.

Uniform water. The axis cavitates first, at the swirl where its pressure deficit reaches $p_\infty - p_{\mathrm{sat}} + \Delta_n$. For a core of the Lamb–Oseen form that swirl is $\sqrt{(p_\infty - p_{\mathrm{sat}} + \Delta_n)/(1.702\,\rho)}$. It rises with the pressure the chamber is retained at and falls as the gas dissolved in the water rises. No heating is measurable before it: 0.06 kelvin at 17 meters per second.

Water in layers. The author’s addition to I13 is scored a hit on the chemical difference. Pimenova and Goldobin [19] report water and n-heptane boiling at their interface at 78.56 °C while neither bulk boils, and Pfeiffer and others [18] report bubbles under tension nucleating mostly along the interface between water and a perfluorocarbon that holds more gas. A column carrying such a layer along its axis is expected to cavitate at a lower swirl than uniform water, and at the interface first. A layer of warmer water lowers the swirl through p_{sat} alone: on the IAPWS values, 7.46 meters per second at 40 °C, 6.97 at 60 °C and 5.71 at 80 °C, against 7.63 at 20 °C, before any change in Δ_n. A difference of density alone has no source obtained, and nothing is predicted for it.

What would count against it. Vapor appearing in the bulk of uniform water away from the axis before the axis cavitates; heating of a kelvin or more before the onset; or a column with such a layer that cavitates at the same swirl as uniform water, or in the bulk before the interface.

Status: theory. The figures are computed from exact solutions and the IAPWS formulations, and the expectations for layers are read off the two sources named. No such chamber has been built or run for this work.

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